1.Write a program that implements the Concept of Encapsulation.

Source Code: -

```
class Student {
    private int id;
    private String name;
    public void setData(int i, String n) {
        id = i;
        name = n;
    }
    public void showData() {
System.out.println("ID: " + id);
System.out.println("Name: " + name);
    }
}
public class Encapsulation {
    public static void main(String[] args) {
        Student s = new Student();
s.setData(101, "jharna");
s.showData();
    }
}
```

Output:-

```
run:
ID: 101
Name: jharna
BUILD SUCCESSFUL (total time: 0 seconds)
```

2.Write a program to demonstrate concept of Method Overloading.

Source Code: -

```
package overpoly;
class Overload {
   int add(int a, int b) {
      return a + b;
   }

   int add(int a, int b, int c) {
      return a + b + c;
   }
}

public class FunctionOverloading {
   public static void main(String[] args) {
      Overload o = new Overload();
System.out.println(o.add(5, 10));
System.out.println(o.add(1, 2, 3));
   }
}
```

Output:-

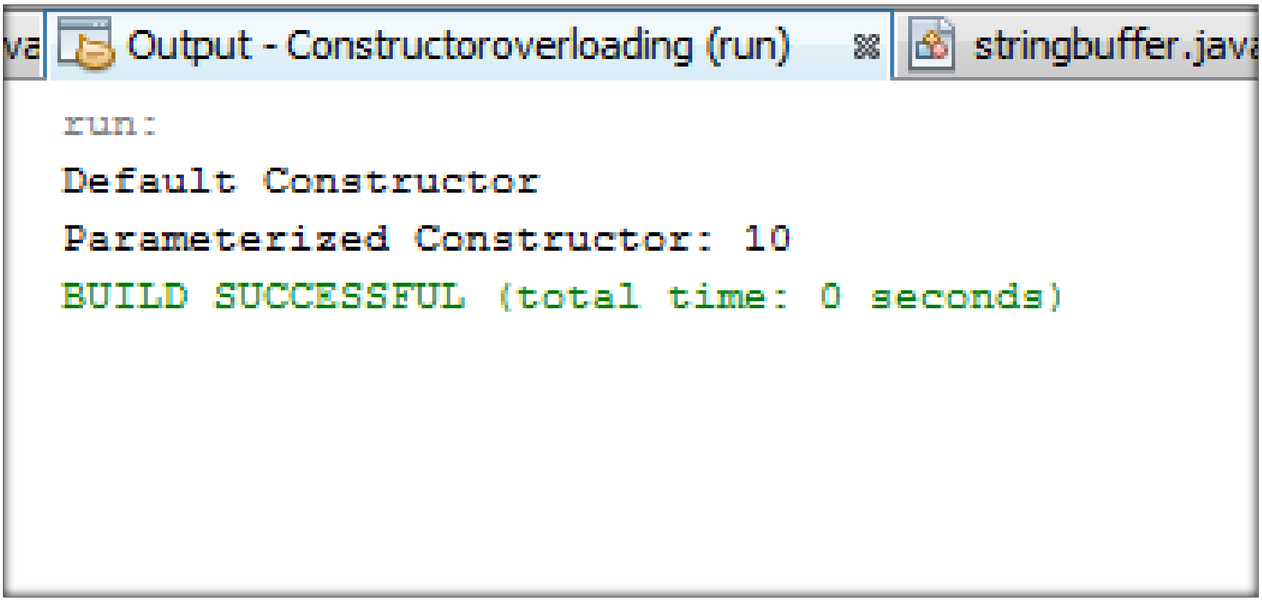
```
run:
15
6
BUILD SUCCESSFUL (total time: 0 seconds)
```

3. Write a program to demonstrate concept of constructor overloading.

Source Code: -

```
package constructoroverloading;
class Demo {
Demo() {
System.out.println("Default Constructor");
   }
Demo(int a) {
System.out.println("Parameterized Constructor: " + a);
   }
}
public class ConstructorOverloading {
   public static void main(String[] args) {
      new Demo();
      new Demo(10);
   }}
```

Output:-



4. Write a program to print the Prime Number Series up to 50.

Source Code: -

```
package primeseriess;
public class Primeseriess {
   public static void main(String[] args) {
       for (int i = 2; i <= 50; i++) {
                boolean prime = true;
          for (int j = 2; j < i; j++) {
             if (i % j == 0) {
                prime = false;
                break;
             }
          }
          if (prime)
             System.out.print(i + " ");
       }}}
```

Output:-

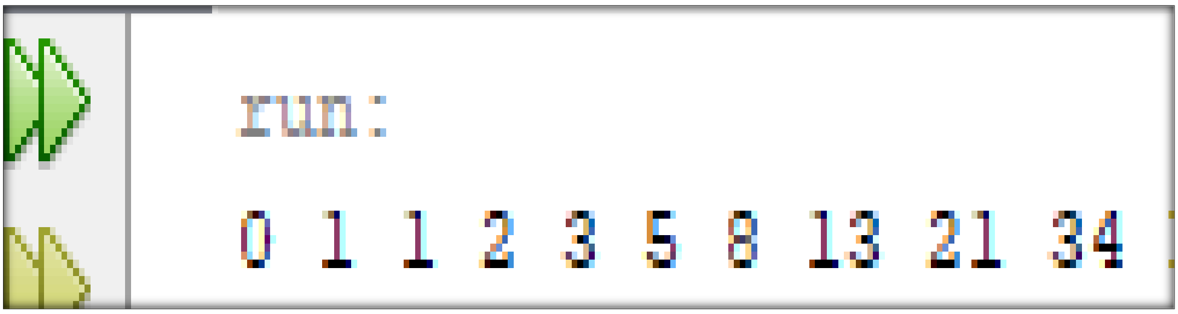
```
run:
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 BUILD SUCCESSFUL (total time: 0 seconds)
```

5. Write a program to print first 10 number of the following Series using Do-While Loop 0, 1, 1, 2, 3, 5, 8, 13...

<u>Source Code: -</u>

```
package javaapplication17;
public class Ass5 {
 public static void main(String[] args) {
  int a = 0, b = 1, c, i = 1;
   do {
          System.out.print(a + " ");
           c = a + b;
           a = b;
           b = c;
           i++;
        } while (i <= 10);
}}
```

<u>Output:-</u>



6. Write a program to check the given number is Armstrong or not.

Source Code: -

```
package armstrong2;
import java.util.Scanner;
public class Armstrong2 {
 public static void main(String[] args) {
  Scanner sc = new Scanner(System.in);
   System.out.println(“Enter any number:”)
   int n = sc.nextInt();
        int sum = 0, temp = n;
        while (n > 0) {
            int r = n % 10;
            sum += r * r * r;
            n /= 10;
        }

if (sum == temp)
 System.out.println("Armstrong Number");
    else
      System.out.println("Not Armstrong");
    }
}
```

Output:-

```
run:
Enter any number
153
Armstrong Number
BUILD SUCCESSFUL (total time: 4 seconds)
```

7. Write a program to find the factorial of any given number.

Source Code: -

```
package factorial;
import java.util.*;

public class jharnaFactorial {
   public static void main(String[] args) {
      int fact =1, n, i;
      Scanner obj=new Scanner(System.in);
      System.out.println("Enter value of n:");
      n=obj.nextInt();
      for(i=1;i<=n;i++)
      {
         fact=fact*i;
      }
      System.out.println("factorial="+fact);
   }}
```

Output:-

```
run:
Enter value of n:

4
factorial=24
BUILD SUCCESSFUL (total time: 21 seconds)
```

8. Write a program to sort the element of One Dimensional Array in Ascending order.

Source Code: -

```
package  sortarray;
import java.util.Scanner;
public class SortArray {
 public static void main(String[] args) {
 Scanner sc=new Scanner(System.in);
 int[] a=new int[5];
 System.out.println("Enter elements");
 for(int i=0;i<5;i++)
  a[i]=sc.nextInt();
  for(int i=0;i<5;i++)
  for(int j=i+1;j<5;j++)
   if(a[i]>a[j]) {
    int t=a[i]; a[i]=a[j]; a[j]=t;
     }
for(int x:a) System.out.print(x+" ");
    }
}
```

Output:-

```
run:
Enter elements
6
4
3
2
4
2 3 4 4 6 BUILD SUCCESSFUL
```

9. Write a program for matrix multiplication using input/output Stream.

Source Code: -

```
package multimatrix;
import java.util.*;
public class jharnaMultimatrix {
 public static void main(String[] args) {
 int a[][]=new int [2][2];
  int b[][]=new int[2][2];
   int c[][]=new int[2][2];
   int i,j,k;
  Scanner sc = new Scanner(System.in);
   System.out.println("enter array elements");
     for(i=0;i<2;i++)
    {
      for(j=0;j<2;j++)
     {
       a[i][j]=sc.nextInt();
      }      }
      System.out.println("Array elements are");
        for(i=0;i<2;i++)
        {
         for(j=0;j<2;j++)
            {
             System.out.println(a[i][j]);
          }
        System.out.println();
        }
        System.out.println("enter array elements");
        for(i=0;i<2;i++)
        {
           for(j=0;j<2;j++)
           {
              b[i][j]=sc.nextInt();
           }               }
        System.out.println("Array elements are");
        for(i=0;i<2;i++)
        {
           for(j=0;j<2;j++)
           {
              System.out.println(b[i][j]);
         }
           System.out.println();
        }
        System.out.println("multiplication of matrix");
        for(i=0;i<2;i++)
        {
           for(j=0;j<2;j++)
```

```
    {
       c[i][j]=0;
       for(k=0;k<2;k++)
  {
   c[i][j]=c[i][j]+a[i][k]*b[k][j];
  }       }       }
for(i=0;i<2;i++)
{
   for(j=0;j<2;j++)
    {
System.out.println(c[i][j] +"");
    }
System.out.println();
}    }   }
```

Output:-

```
run:
enter array elements
1
2
3
4
Array elements are
1
2

3
4

enter array elements
6
7
8
9
Array elements are
6
7

8
9

multiplication of matrix
22
25

50
57

BUILD SUCCESSFUL (total time: 10 seconds)
```

10. Write a program for matrix addition using input/output stream class.

Source Code: -

```
package matrixadd;
import java.util.*;
public class jharnaMatrixadd {
 public static void main(String[] args) {
       int a[][]=new int[2][2];
        int b[][]=new int[2][2];
         int c[][]=new int[2][2];
  Scanner a1 = new Scanner(System.in);
   System.out.println("enter 1st  array elements:");
   for(int i=0;i<2;i++){
           for(int j=0;j<2;j++){
              a[i][j]=a1.nextInt();
           }          }
   System.out.println("enter 2nd  array elements:");
         for(int i=0;i<2;i++){
           for(int j=0;j<2;j++){
              b[i][j]=a1.nextInt();
           }  }
      for(int i=0;i<2;i++){
            for(int j=0;j<2;j++){
         c[i][j]=a[i][j]+b[i][j];
            }}    System.out.println("Addition is: - ");
    for(int i=0;i<2;i++){
           for(int j=0;j<2;j++)
           {           System.out.print(c[i][j]+" ") ;
            }
            System.out.println("");
           }    }    }
```

Output:-

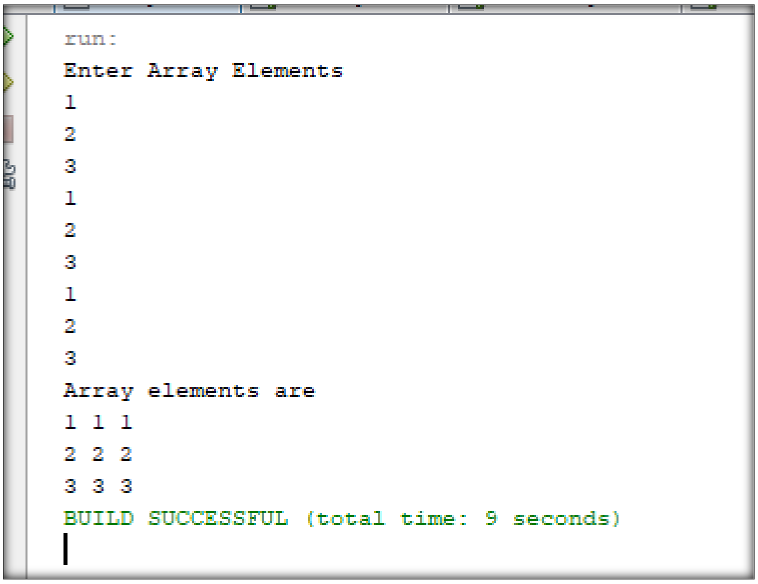
```
run:
enter 1st array elements:
12
23
34
45
enter 2nd array elements:
54
43
32
21
Addition is :-
66 66
66 66
BUILD SUCCESSFUL (total time: 9 seconds)
```

11. Write a program for matrix transpose using input/output stream class.

Source Code: -

```
package transpose;
import java.util.*;
public class Transpose {
public static void main(String[] args) {
    int a[][]=new int[3][3];
    int i,j;
    Scanner s1=new Scanner(System.in);
    System.out.println("Enter Array Elements");
    for(i=0;i<3;i++)
    {for(j=0;j<3;j++)
    {a[i][j]=s1.nextInt();
       }
    }
    System.out.println("Array elements are");
    for(i=0;i<3;i++)
    {for(j=0;j<3;j++)
    {System.out.print(a[j][i]+ " ");
       } System.out.println();
} }}
```

Output:-



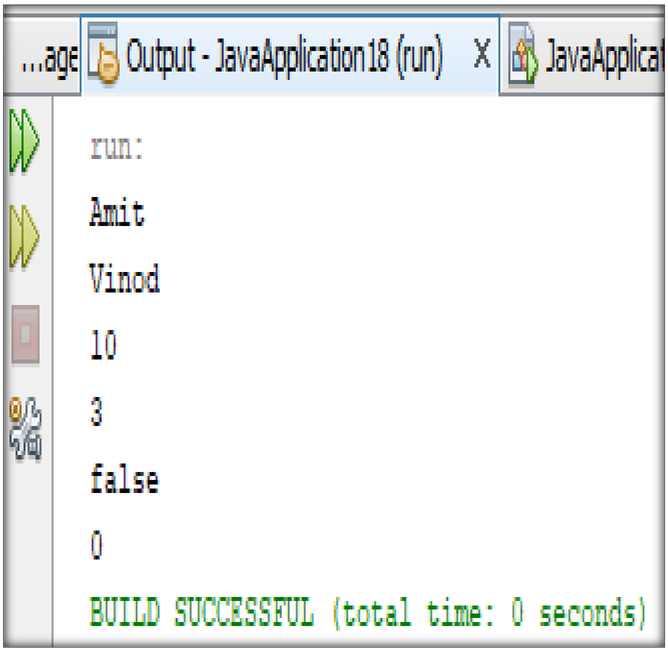
```
run:
Enter Array Elements
1
2
3
1
2
3
1
2
3
Array elements are
1 1 1
2 2 2
3 3 3
BUILD SUCCESSFUL (total time: 9 seconds)
```

12. Write a program for Vector class which performs all the methods of that class.

<u>Source Code: -</u>

```
package javaapplication18;
import java.util.Vector;
public class JavaApplication18 {
 public static void main(String[] args) {
        Vector v = new Vector();
      v.addElement("Sagar");
      v.addElement("Amit");
       v.addElement("Vinod");
       for(int i=1;i<3;i++)
       {
 System.out.println(v.elementAt(i));
       }
      System.out.println(v.capacity());
    System.out.println(v.size());
    System.out.println(v.isEmpty());
         v.clear();
      System.out.println(v.size());
    }}
```

<u>Output:-</u>



13. Write a program to check that the given String is palindrome or not.

Source Code: -

```
package javaapplication17;
import java.util.Scanner;

public class NewMain {
   public static void main(String[] args) {
   Scanner sc = new Scanner(System.in);
 System.out.print("Enter a string: ");
      String str = sc.nextLine();
      String rev = "";
         for (int i = str.length() - 1; i >= 0; i--) {
         rev = rev + str.charAt(i);
      }
      if (str.equals(rev)) {
         System.out.println("The string is a Palindrome");
      } else {
         System.out.println("The string is NOT a Palindrome");
      }}}
```

Output:-

```
run:
Enter a string: madam
The string is a Palindrome
BUILD SUCCESSFUL (total time: 5 seconds)
```

14. Write a program to arrange the String in alphabetical order.

Source Code: -

```
package muskan;
import java.util.Arrays;
import java.util.Scanner;
public class ass14 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String input = sc.nextLine();
        char[] chars = input.toCharArray();
                Arrays.sort(chars);
        String sortedString = new String(chars);
        System.out.println("String in alphabetical order: " + sortedString);
        sc.close();
    }
}
```

Output:-

```
run:
Enter a string: tomorrow
String in alphabetical order: mooorrtw
BUILD SUCCESSFUL (total time: 47 seconds)
```

15.Write a program for String class which performs all the methods of that class.

Source Code: -

```
package javaapplication17;
import java.util.*;
public class Assi15 {
    public static void main(String[] args) {
     String str1,str2;
     Scanner A = new Scanner(System.in);
     str1="Pallotti";
     str2="College";
     System.out.println(str1.length());
    System.out.println(str1.toUpperCase());
    System.out.println(str1.toLowerCase());
    System.out.println(str1.concat(str2));
    System.out.println(str1.equals(str2));
    System.out.println(str1.charAt(2));
      System.out.println(str1.replace('0','a'));
       System.out.println(str1.contains("Pal"));
        System.out.println(str2.trim());
         System.out.println(str1.equalsIgnoreCase(str2));
  }   }
```

Output:-

```
run:
8
PALLOTTI
pallotti
PallottiCollege
false
l
Pallotti
true
College
false
BUILD SUCCESSFUL (total time: 0 seconds)
```

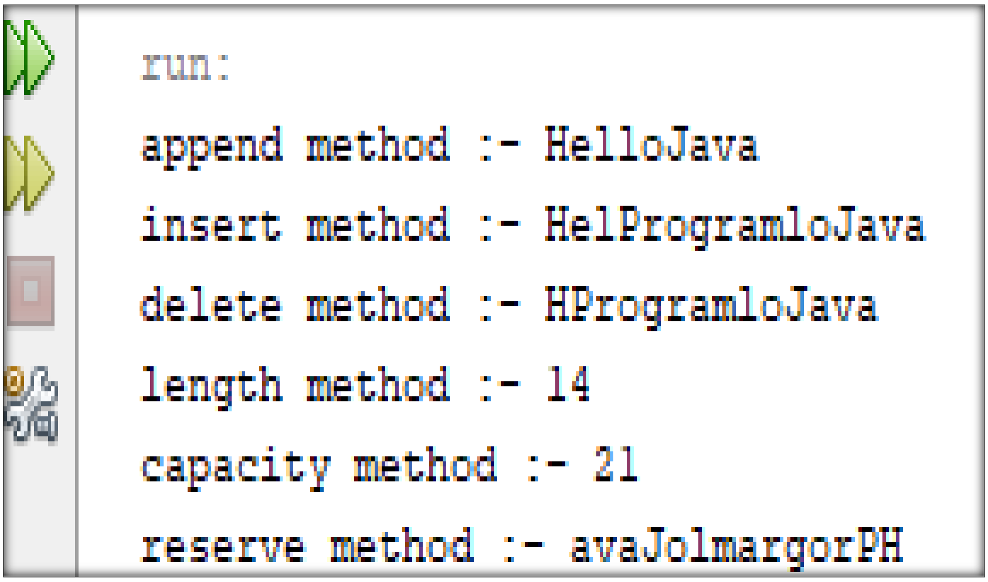
16. Write a program for StringBuffer class which perform the all methods of that class.

Source Code: -

```
package javaapplication17;
import java.util.*;
public class Assi16 {
public static void main(String[] args) {
     StringBuffer s1 = new StringBuffer ("Hello");

     System.out.println("append method :- "+ s1.append("Java"));
     System.out.println("insert method :- "+ s1.insert(3,"Program"));
     System.out.println("delete method :- "+ s1.delete(1,3));
     System.out.println("length method :- "+ s1.length());
     System.out.println("capacity method :- "+ s1.capacity());
     System.out.println("reserve method :- "+ s1.reverse());
    }}
```

Output:-



17. Write a program to calculate Simple Interest using the Wrapper Class.

Source Code: -

```
package SimpleInt;
public class SimpleInterest {
    public static void main(String[] args) {
        Integer p=1000, r=5, t=2;
        int si=(p*r*t)/100;
System.out.println("SI = "+si);
    }
}
```

Output:-

```
run:
SI = 100
BUILD SUCCESSFUL (total time: 0 seconds)
```

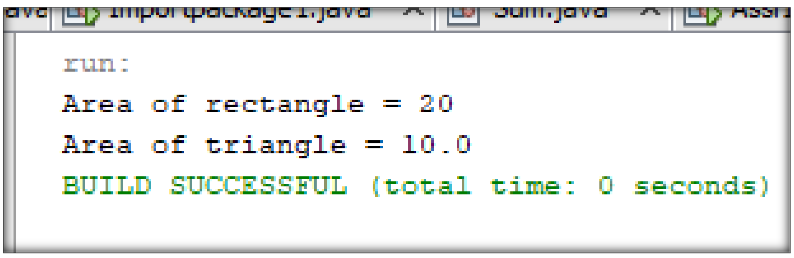
18. Write a program to calculate area of various geometrical figures using the abstract class.

Source Code: -

```
package javaapplication17;
abstract class Shape {
   abstract void area();
   }
class Rect extends Shape {
   int length, breadth;

   Rect(int l, int b) {
      length = l;
      breadth = b;
   }
   void area() {
      System.out.println("Area of rectangle = " + (length * breadth));
   }   }
class tri extends Shape
{
int base,height;
tri(int b,int h)
{
   base=b;
   height=h;
}
void area()
{
 System.out.println("Area of triangle = " + (0.5f*base * height));
}}
public class Assi18 {
    public static void main(String[] args) {
         Shape s = new Rect(5, 4);
         Shape s1 = new tri(5, 4);
         s.area();
         s1.area();
      }  }
```

Output:-



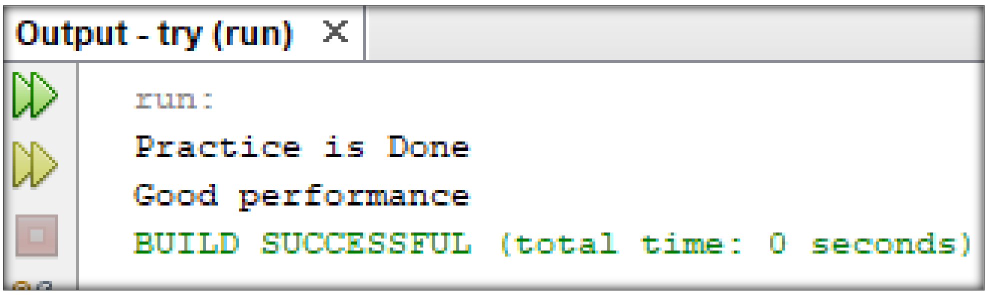
19. Write a program where Single class implements more than one interface.

Source Code: -

```
package pkgtry;
interface  sports
{
void practice();
}
interface participants
{
void performance();
}
class coach implements sports, participants
{
public void practice()
{
System.out.println("Practice is Done");
}
 public void performance()
{
System.out.println("Good performance ");
}
}
public class Try {
public static void main(String[] args) {
       coach c = new coach();
       c.practice();
       c.performance();
    }

}
```

Output:-



20. Write a program that use the multiple catch statements within the try-catch mechanism.

<u>Source Code: -</u>

```
package multiplecatchdemo;
import java.util.Scanner;
public class MultipleCatchDemo {
  public static void main(String[] args) {
     Scanner sc = new Scanner(System.in);
 try {
            System.out.print("Enter first number: ");
            int a = sc.nextInt();
            System.out.print("Enter second number: ");
            int b = sc.nextInt();
            int c = a / b;
            System.out.println("Result = " + c);
       }
 catch (ArithmeticException e) {
            System.out.println("Error: Division by zero is not allowed");
          } catch (NumberFormatException e) {
            System.out.println("Error: Invalid number format");
         } catch (Exception e) {
            System.out.println("General Exception Occurred");
        }
System.out.println("Program continues...");
  } }
```

<u>Output:-</u>

```
run:
Enter first number: 1
Enter second number: 0
Error: Division by zero is not allowed
Program continues...
BUILD SUCCESSFUL (total time: 4 seconds)
```

21. Write a program where user will create a self-Exception using the "throw" keyword.

Source Code: -

```
package userdefinedexceptiondemo;
import java.util.Scanner;
class InvalidAgeException extends Exception {
    InvalidAgeException(String msg) {
        super(msg);
    }}
public class UserDefinedExceptionDemo {
 public static void main(String[] args) {
                Scanner sc = new Scanner(System.in);
 try {
            System.out.print("Enter your age: ");
            int age = sc.nextInt();
          if (age < 18) {
                // throw user-defined exception
                throw new InvalidAgeException("Age must be 18 or above");
            }
        System.out.println("You are eligible.");
          } catch (InvalidAgeException e) {
            System.out.println("Exception Caught: " + e.getMessage());
        }    }   }
```

Output:-

```
run:
Enter your age: 15
Exception Caught: Age must be 18 or above
BUILD SUCCESSFUL (total time: 3 seconds)
```

22. Write a program for multithread using different methods of Thread class.

Source Code: -

```
package javaapplication8;
class MyThread1 extends Thread {
   public void run() {
   System.out.println(getName() + " is running");
   for (int i = 1; i <= 3; i++) {
         System.out.println(getName() + " → " + i);
         try {
            Thread.sleep(500);
         } catch (Exception e) {}
      }
  System.out.println(getName() + " finished");
   }}
  public class threadmethods {
  public static void main(String[] args) {
    MyThread1 t1 = new MyThread1();
    MyThread1 t2 = new MyThread1();
   t1.setName("Thread A");
   t2.setName("Thread B");
   t1.setPriority(3);
   t2.setPriority(8);
    System.out.println(t1.getName() + " alive? " + t1.isAlive());
    t1.start();
    t2.start();
   System.out.println(t1.getName() + " alive? " + t1.isAlive());
    try {           t1.join();
    t2.join();
    } catch (Exception e) {}
     System.out.println(t1.getName() + " alive? " + t1.isAlive());
     System.out.println("Main thread finished");
   }}
```

Output:-

```
run:
Thread A alive? false
Thread A alive? true
Thread B is running
Thread B → 1
Thread A is running
Thread A → 1
Thread B → 2
Thread A → 2
Thread A → 3
Thread B → 3
Thread A finished
Thread B finished
Thread A alive? false
Main thread finished
BUILD SUCCESSFUL (total time: 2 seconds)
```

23. Write a program for multithreading using synchronized () method of Thread class.

Source Code: -

```
package javaapplication83;
class table
{
 synchronized void Ptable(int n)
  {
        for(int i=1;i<=10;i++)
        {
           System.out.println(n+"X"+i+"="+(n*i));
           try
           {
              Thread.sleep(3000);
              }
           catch(Exception e)
           {
            System.out.println(e);
           }}}}
class mythread extends Thread{
        table t;
        mythread (table t)
        {
           this.t=t;
        }
        public void run()
        {
           t.Ptable(5);
        }
        }
class mythread2 extends Thread{
    table t;
    mythread2(table t)
    {
        this.t=t;
              }
    public void run()
    {
        t.Ptable(10);
    }
    }
public class synchrojharna{


    public static void main(String[] args) {
        table obj=new table();
        mythread t1=new mythread(obj);
        mythread2 t2 =new mythread2(obj);
```

```
        t1.start();
         t2.start();

    }
}
```

Output:-

```
run:
5X1=5
5X2=10

5X3=15

5X4=20

5X5=25

5X6=30
5X7=35
5X8=40
5X9=45
5X10=50
10X1=10
10X2=20
10X3=30
10X4=40
10X5=50
10X6=60
10X7=70
10X8=80
10X9=90
10X10=100
BUILD SUCCESSFUL (total time: 1 minute 0 seconds)
```

24. Write a program for multithread for setting the priority of the thread and the print their Priority.

Source Code: -

```
package javaapplication17;
class Thread11 extends Thread {
    public Thread11(String name) {
        super(name);
    }
 public void run() {
 for (int i = 1; i <= 5; i++) {
  System.out.println(getName() + " is running with priority " + getPriority());
        }
    }
}
public class Assi24 {
    public static void main(String[] args) {
        Thread1 t1 = new Thread1("High Priority Thread");
        Thread1 t2 = new Thread1("Normal Priority Thread");
        Thread1 t3 = new Thread1("Low Priority Thread");
 t1.setPriority(Thread.MAX_PRIORITY);
        t2.setPriority(Thread.NORM_PRIORITY);
        t3.setPriority(Thread.MIN_PRIORITY);
 t1.start();
        t2.start();
        t3.start();
    }}
```

Output:-

```
run:
High Priority Threadis running with priority10
Low Priority Threadis running with priority1
Normal Priority Threadis running with priority5
Low Priority Threadis running with priority1
High Priority Threadis running with priority10
Low Priority Threadis running with priority1
Normal Priority Threadis running with priority5
Low Priority Threadis running with priority1
High Priority Threadis running with priority10
Low Priority Threadis running with priority1
Normal Priority Threadis running with priority5
High Priority Threadis running with priority10
High Priority Threadis running with priority10
Normal Priority Threadis running with priority5
Normal Priority Threadis running with priority5
BUILD SUCCESSFUL (total time: 0 seconds)
```

25. Write a program to create a package using command and one package will import another package.

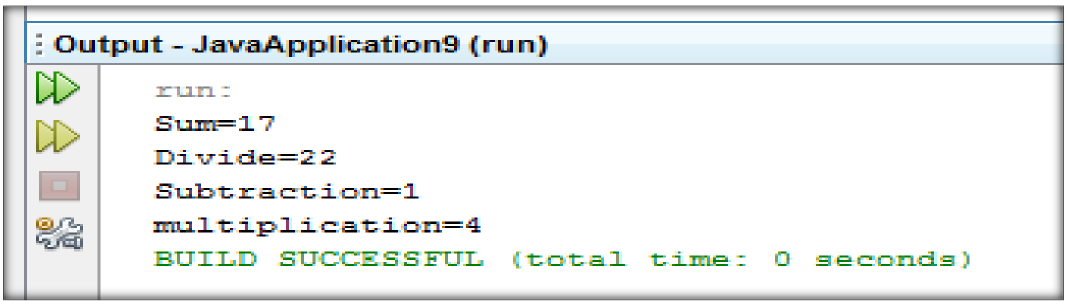
Source Code: -

```
package newpackage;
public class Sum {
   public int a1;
   public  int b;
   public int c;
   int d;
   public int Add(int a,int b)
   { return a+b;
   }     public int Sub(int a,int b)
   { return a-b;
   }   public int Mul(int a,int b)
   { return a*b;
   }     public int Div(int a,int b)
   { return a/b;
   }}
```

Importing user defined Package (mypackage)

```
package importpackage1;
import newpackage.*;
public class Importpackage1 {
public static void main(String[] args) {
Sum a = new Sum();
int add, d, m, s1;
 d = a.Div(20, 2);
 s1 = a.Sub(4, 3);
  m = a.Mul(2, 2);
   add = a.Add(8, 9);
   System.out.println("Sum=" + add);
    System.out.println("Divide=" + d);
    System.out.println("Subtraction=" + s1);
    System.out.println("Multiplication=" + m);
 }}
```

Output:-

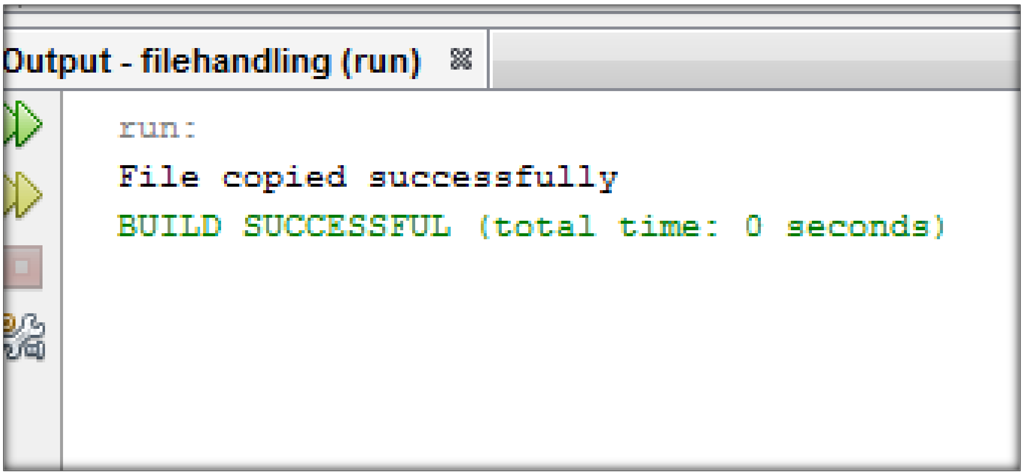


26. Write a program for reading and writing bytes to a file.

Source Code: -

```
 package filehandling;
import java.io.*;
public class Bytestream {
 public static void main(String[] args) {
    try {
            FileInputStream f1 = new FileInputStream("dog.jpg");
            FileOutputStream f2 = new FileOutputStream("dog1.jpg");
            int data;
            while ((data = f1.read()) != -1) {
                f2.write(data); // write byte
            }
      System.out.println("File copied successfully");
            f1.close();
            f2.close();
        }
 catch (Exception e)
{
    System.out.println(e);
  } }}
```

Output:-



27. Write a program for demonstration of switch statement, continue and break.

Source Code: -

```
package switchcontinuebreak;
import java.util.Scanner;
public class SwitchContinueBreak {
 public static void main(String[] args) {
 Scanner sc = new Scanner(System.in);
for (int i = 1; i <= 3; i++) {
            System.out.print("Enter a number (1-3): ");
            int choice = sc.nextInt();
 switch (choice) {
                case 1:
                    System.out.println("You chose 1");
                    break;
                case 2:
                    System.out.println("You chose 2");
                    continue; // skips the rest and goes to next iteration
                case 3:
                    System.out.println("You chose 3");
                    break;
                default:
                    System.out.println("Invalid choice");
            }
            System.out.println("End of iteration " + i);
        }   }   }
```

Output:-

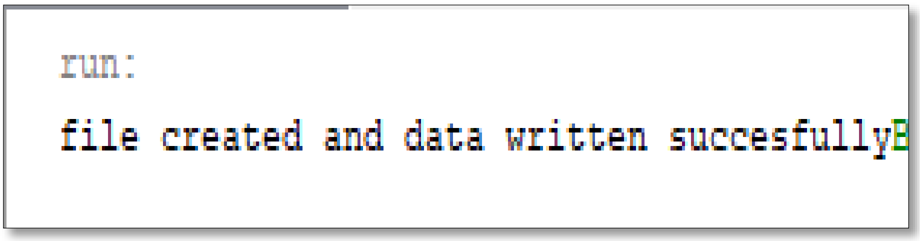
```
run:
Enter a number (1-3): 1
You chose 1
End of iteration 1
Enter a number (1-3): 3
You chose 3
End of iteration 2
Enter a number (1-3): 5
Invalid choice
End of iteration 3
BUILD SUCCESSFUL (total time: 8 seconds)
```

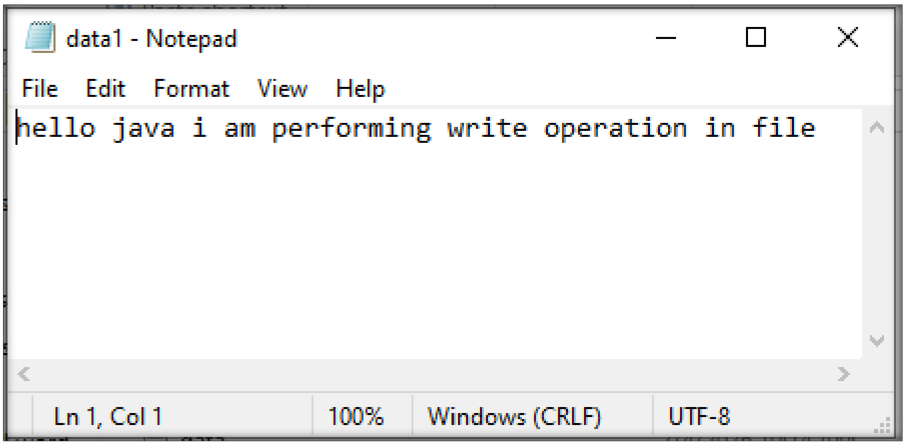
28. Write a program for creating a text file and perform write operation using FileWriter class.

Source Code: -

```
package javaapplication86;
import java.io.*;
public class writerjharna{
  public static void main(String[] args) {
        try{
              FileWriter f =new FileWriter("data1.txt");
              f.write("hello java i am performing write operation in file");
              f.close();
              System.out.print("file created and data written succesfully");
             }
        catch(IOException e)
        {
            System.out.print(e);
        } }}
```

Output:-





29. Write a program for perform read operation using FileReader class.

<u>Source Code: -</u>

```
package javaapplication17;
import java.io.*;
public class Ass29 {
  public static void main(String[] args) {
        try
        {
        FileReader r = new FileReader ("data.txt");
        int ch;
        while((ch=r.read())!=-1)
        {
        System.out.println((char)ch);
        }
        r.close();
        }
        catch(IOException e)
        {
        System.out.println(e);
        }
    }
   }
```

<u>Output:-</u>

```
run:
Hello Java I am performing write operation in file
```

30. Write a program for JDBC to display the records from the existing table

Source Code: -

```
 package connectivity;
import java.sql.*;
public class Connectivity {
 public static void main(String[] args) {
       try{
           Class.forName("sun.jdbc.odbc.JdbcOdbcDriver");
           Connection c=DriverManager.getConnection("jdbc:odbc:db1");
           Statement st=c.createStatement();
           ResultSet rs=st.executeQuery("select *from teacher");
           while(rs.next())
           {System.out.println(rs.getInt(1)+" "+rs.getString(2));
           }
       }
    catch(Exception e)
    {
           System.out.print("error");
    }}}
```

Output:-